Arthur Cayley

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Contents

561. On the Geometrical Representation of Cauchy's Theorems of Root-Limitation (continued)

[Continued from earlier pages of the same paper. Articles 27 (end) through 38, and Memoir list.]

[p. 31] (continuation of the table of intercalations from the previous page):

Interval $0-4$	for P gain $=0$, … for Q gain $=-1$;	Intercalation is Q;
$0-4$	“ $=-1$, Q first,	“ $=-1$; “ QP;
$2-4$	“ $=1$, P first,	
$2-4$	“ $=-1$, P first,	“ $=0$; “ P.

28. Or to take a slightly more complicated example,

		1	3	5	7	9
$P=x^2-8x+12$	+	−	−	+	+	+
$Q=x^2-12x+32$	+	+	−	−	−	+
$R=-x+5$	+	+	+	+	−	−
$S=+1$	+	+	+	+	+	+
Showing P	P		Q		P	
	1	2	3	4		

And hence for the several intervals,

	1−3	1−5	1−7	1−9	3−5	3−7	3−9	5−7	5−9	7−9
	−−	−+	−+	−−	−+	−+	−−	−−	−+	−+
	+−	+−	++	++	+−	+−	+−	++	+−	++
	++	+−	++	+−	+−	++	++	+−	+−	+−
	++	++	++	++	++	++	++	++	++	++
Showing P	PQ	PQP	$PQPQ$	Q	QP	QPQ	P	PQ	Q	

For instance:
Interval $1-9$ for P gain $=2$, P first, for Q gain $=2$: Intercalation is $PQPQ$.
It may be added that P being $+$ for $x=1$, the $\pm$ intercalation is $+PQPQ$.

29. As an example of circuits take the following: curves are $P=0$, $Q=0$, where

$$P=x^2+y^2-4,$$

$$Q=y-x-1;$$

viz. $P=0$ (see figure) is a circle, centre the origin, radius $=2$: the inside hereof ($P=-$) being coloured red: and $Q=0$ is a right line cutting the axes of x, y at the points $(-1,0)$ and $(0,1)$ respectively, or say running N.E. and S.W., the lower region ($Q=-$) being coloured blue: the square is an arbitrary circuit ($x=\pm3$, $y=\pm3$) surrounding the circle, and the regions within the square are coloured by what precedes sable, gules, azure, argent, as shown in the figure: the line and circle intersect in two points M, N. Going right-handedly round these respectively, for M the order is sable, gules, argent, azure, viz. M is a right-handed root; while for N the order is

[p. 32] sable, azure, argent, gules, viz. N is a left-handed root: the two points are accordingly in the figure denoted by $+M$ and $-N$ respectively.

[Figure: a square of side 6, centre at origin, containing a circle of radius 2; the diagonal line $y=x+1$ crosses the circle at M (upper-right) and N (lower-left); the four quadrants of the square outside the circle are tinted sable, gules, azure, argent.]

30. Now considering successively the four smaller squares of the figure, say these are the squares N.E., S.E., S.W., N.W.: and going right-handedly round each of these—

In the square N.E., the sequence and therefore also the intercalation is $+P-Q+P-Q$, viz. this is an intercalation $(+P-Q)_1$, showing an excess 1 of right-handed roots, and of course consisting with the single right-handed root M.

In the square S.E., the sequence is $-P+P$, viz. this is an intercalation $(PQ)_0$, showing an equality of right- and left-handed roots, and consisting with no root.

In the square S.W., the sequence and therefore also the intercalation is $-P+Q-P+Q$: viz. this is an intercalation $(-P+Q)_1$, showing an excess 1 of left-handed roots, and consisting with the single left-handed root N.

And in the square N.W., the sequence is $-Q+P-P+Q$, viz. this is an intercalation $(PQ)_0$, showing an equality of right- and left-handed roots, and consisting with no root.

Again take the whole large square: the sequence is $-Q+Q$: viz. the intercalation is $(PQ)_0$, showing an equality of right- and left-handed roots, and consisting with there being one of each.

So taking the squares N.E. and N.W. conjointly, the sequence and therefore also the intercalation is $-Q+P-Q+P$, viz. this is an intercalation $(+P-Q)_1$, as for the single square N.E.

[p. 33] **31.** As regards the analytical determination it will be sufficient to consider a single square, say N.E.: going round right-handedly, the trajectories will be

$$\begin{aligned}
&(1)\ x=0,\ \ y=0 \text{ to } 3\\
&(2)\ y=3,\ \ x=0 \text{ to } 3\\
&(3)\ x=3,\ \ y=3 \text{ to } 0\,;\ \text{or if } y'=-y,\ \text{then } y'=-3 \text{ to } 0\,;\\
&(4)\ y=0,\ \ x=3 \text{ to } 0\,;\ \text{or if } x'=-x,\ \text{then } x'=-3 \text{ to } 0.
\end{aligned}$$

And we thus have

		0	3	
(1) $P=y^2-4$	$-$	$+$	$+$	that is, $-+$ for P gain $=-1$, Q begins, $1P$
$Q=y-1$	$-$	$-$	$+$	Q $=-1$, $1Q$.
$R=-1$	$-$	$-$	$-$	

Intercalation is QP, or since at origin $P=-$, $R=-$, or region is sable, it is $-Q+P$.

		0	3	
(2) $P=x^2+5$	$+$	$+$	$+$	that is, $-+$ for P gain $=0$,
$Q=-x+2$	$+$	$+$	$-$	Q $=-1$.
$R=-1$	$-$	$-$	$-$	

Intercalation is $-Q$.

		-3	-3	
(3) $P=y'^2+5$	$+$	$+$	$+$	that is, $--$ for P gain $=0$,
$Q=-y'+4$	$+$	$+$	$+$	Q $=0$.
$R=-1$	$-$	$-$	$-$	

Intercalation vanishes.

		-3	-3	
(4) $P=x'^2-4$	$+$	$-$	$-$	that is, $--$ for P gain $=+1$, P first,
$Q=x'-1$	$-$	$-$	$-$	Q $=0$.

Intercalation is $+P$.

Hence for the four sides, combining the intercalations, we have $-Q+P-Q+P$, and since there are no terms to be omitted, this is the intercalation of the N.E. square: which is right.

C. IX.

[p. 34]

The Rhizic Theory. Articles Nos. 32 to 38.

32. Consider now $F(z) = (*)(z,1)^n$ a rational and integral function of z, of the order n with in general imaginary (complex) coefficients, or, what is the same thing, let $F(z) = f(z) + i\phi(z)$, where the functions f, ϕ are real[1]. Writing herein $z = x + iy$, let P, Q be the real part and the coefficient of the imaginary part in the function $F(x+iy)$: or, what is the same thing, assume

$$P + iQ = f(x+iy) + i\phi(x+iy),$$

then it is clear that to any root $\alpha + i\beta$ (real or imaginary) of the equation $F(z) = 0$, there corresponds a real intersection, or root, $x = \alpha$, $y = \beta$, of the curves $P = 0$, $Q = 0$. The functions P, Q, as thus serving for the determination of the roots of the equation $F(z) = 0$, are termed "rhizic functions," and similarly the curves $P = 0$, $Q = 0$ are "rhizic curves." The assumed equation shows at once that we have

$$d_y(P+iQ) = i\, d_x(P+iQ),$$

or, what is the same thing,

$$d_yP = -d_xQ, \qquad d_xP = d_yQ.$$

And we hence see that

$$\frac{d(P,Q)}{d(x,y)} = (d_xP)^2 + (d_yP)^2, \text{ or } (d_xQ)^2 + (d_yQ)^2,$$

is positive: viz. that the roots $P = 0$, $Q = 0$ are all of them right-handed (the essential thing is that they are same-handed; for by reversing the signs of P and Q they might be made left-handed: but it is convenient to take them as right-handed)—hence the theorem which in the general case, where P and Q are arbitrary functions, serves to determine the difference of the numbers of the right- and left-handed roots, in the particular case, where P and Q are rhizic functions, serves to determine the number of intersections of the curves $P = 0$, $Q = 0$: or, what is the same thing, the number of the (real or imaginary) roots of the equation $F(z) = 0$: viz. we thus determine the number of roots within a given circuit.

33. The rhizic curves $P = 0$, $Q = 0$ have various properties. 1°. Each curve has n real points at infinity, or, what is the same thing, n real asymptotes: and the P and Q points at infinity succeed each other, a P-point and then a Q-point, and so on, alternately.

In fact, from the equation

$$P + iQ = (a' + ia'')(x+iy)^n + \ldots + (k' + k''i),$$

writing herein $a' + ia'' = \alpha(\cos a + i \sin a)$, and $x + iy = \rho(\cos\theta + i\sin\theta)$, we have

$$P + iQ = \alpha\rho^n\left[\cos(n\theta + a) + i\sin(n\theta + a)\right] + \ldots + k' + k''i.$$

[p. 35] It thus appears that for the curve $P = 0$, the points at infinity are given by the equation $\cos(n\theta + a) = 0$; while for the curve $Q = 0$, they are given by the equation $\sin(n\theta + a) = 0$: which proves the theorem.

Representing infinity as a closed curve or circuit, each point at infinity must be represented by two opposite points on the circuit; so that writing down P for each P-point and Q for each Q-point, we have $2n$ P's and $2n$ Q's succeeding each other, a P-point and then a Q-point, and so on, alternately.

It may be assumed that taking the circuit right-handedly, the P's are + and the Q's −, (this depends only on the colouring, but it corresponds with the foregoing assumption that the roots $P = 0$, $Q = 0$ are right-handed): the theorem just obtained then really is that for the circuit infinity, the intercalation is $(+P - Q)_n$: and we have herein a proof of the theorem that a numerical equation of the order n with real or imaginary coefficients has precisely n real or imaginary roots. But the force of this will more distinctly appear presently.

34. 2°. Neither of the curves $P = 0$, $Q = 0$ can include as part of itself a closed curve or circuit.

[1] It is assumed that the equation $F(z) = 0$ has no equal roots: this being so, the curves $P = 0$, $Q = 0$, will have no point of multiple intersection; which accords with the assumption made in the general case of two arbitrary curves.

The foregoing relations between the differential coefficients give

$$d_x^2 P + d_y^2 P = 0, \qquad d_x^2 Q + d_y^2 Q = 0,$$

which equations for the two curves respectively lead to the theorem in question. For as regards the curve $P = 0$, take z a coordinate perpendicular to the plane of xy, and consider the surface $z = P$: if the curve $P = 0$ included as part of itself a closed curve, then corresponding to some point (x, y) within the curve we should have z a proper maximum or minimum, viz. there would be a summit or an imit; at the point in question we should have $d_x P = 0$, $d_y P = 0$; and also (as the condition of a summit or imit) $d_x^2 P \cdot d_y^2 P - (d_x d_y P)^2 = +$, implying that $d_x^2 P$ and $d_y^2 P$ have at this point the same sign: but this is inconsistent with the foregoing relation $d_x^2 P + d_y^2 P = 0$.

35. 3°. The curves $P = 0$, $Q = 0$ have not in general any double (or higher multiple) points. A point which is a double (or higher multiple) point on one of these curves is not of necessity a point on the other curve: but being a point on the other curve it is on that curve a point of the same multiplicity. For changing if necessary the coordinates, the point in question may be taken to be at the origin: forming the equation

$$P + iQ = (a' + a''i)(x + iy)^n + \ldots + (l' + l''i)(x + iy)^2 + (m' + m''i)(x + iy) = 0,$$

the point $x = 0$, $y = 0$ will not be a double point on the curve $P = 0$, unless we have $m' = 0$, $l' = 0$, $l'' = 0$; these conditions being satisfied, it will not be a point on the curve $Q = 0$ unless also $m'' = 0$; but this being so, it will be a double point on the curve $Q = 0$: and the like for points of higher multiplicity. But a point which is a multiple point on each curve, represents four or more coincident intersections of the curves $P = 0$, $Q = 0$, that is, four or more equal roots of the equation $F(z) = 0$; so that assuming that the equation has no equal roots, the case does not arise: and we in fact exclude it from consideration.

[p. 36] To fix the ideas assume that the curves $P = 0$, $Q = 0$ are each of them without double points. As already seen, neither of them includes as part of itself a closed curve. Hence in the figure the curve $P = 0$ must consist of n branches each drawn from a point P in the circuit (viz. the circuit infinity) to another point P in the circuit; and in such manner that no two branches intersect each other: this implies that the two points P of the same branch must include between them an even number (which may of course be $= 0$) of points P. And the like as regards the curve $Q = 0$.

36. 4°. No branch of the P-curve can meet a branch of the Q-curve more than once. In fact, in drawing the two branches to meet twice, the colouring would at once show that of the two intersections or roots, one must be right, the other left-handed; whence, the roots being all right-handed, the branches do not meet twice. And in exactly the same way it appears that no P-branch can meet two Q-branches, or any Q-branch meet two P-branches. And under these restrictions it requires only a consideration of a few successive cases to show that the n P-branches, and the n Q-branches can only be drawn on the condition that each P-branch shall intersect once and only once a single Q-branch; which of course implies that each Q-branch intersects once and once only a single P-branch: and further, that there shall be precisely n intersections: viz. the n P-branches and the n Q-branches must satisfy the conditions just stated. And the theorem of the n roots is thus obtained as a consequence of the impossibility (except under the same conditions) of drawing the n P-branches and the n Q-branches, so as to give rise to right-handed roots only. But the case of double or higher multiple points would need to be specially considered.

37. It is interesting for a given value of n to consider $\phi(n)$, the number of different ways in which the P-branches and the Q-branches can be drawn. We have $2n$ points P and $2n$ points Q, in all $4n$ points: starting from any point P, these may be numbered in order $1, 2, 3, \ldots, 4n$, the points P bearing odd numbers and the points Q even numbers. We may consider the P-branch which joins 1 with some P-point β, and (intersecting this) the Q-branch which joins some two Q-points α and γ: the numbers $1, \alpha, \beta, \gamma$ are then in order of increasing magnitude: and excluding these four points there remain the points corresponding to numbers between 1 and α, between α and β, between β and γ, and between γ and 1. Now since the P-branch 1β meets the Q-branch $\alpha\gamma$, no branch from a point between 1 and α can meet either of these curves; hence these points form a system by themselves, capable of being connected together by P-branches and Q-branches: the number of them must therefore be a multiple of 4: and the like as to the points between α and β, between β and γ, and between γ and 1. Taking the number of the points in the four systems to be $4x$, $4y$, $4z$, and $4w$ respectively, we have $x+y+z+w=n-1$, and the first-mentioned four points bear the numbers

$$\begin{aligned}&1,\\ &\alpha = 4x+2,\\ &\beta = 4x+4y+3,\\ &\gamma = 4x+4y+4z+4.\end{aligned}$$

[p. 37] For the four systems the number of ways of drawing the P- and Q-branches are ϕx, ϕy, ϕz, ϕw respectively: that is, x, y, z, w being any partition whatever of $n-1$ (order attended to), and $\phi(0)$ being $=1$, we have

$$\phi(n) = \Sigma\, \phi(x)\, \phi(y)\, \phi(z)\, \phi(w),$$

which is the condition for the determination of ϕn.

Taking then θ for the value of the generating function

$$1 + t\phi(1) + t^2\phi(2) + \ldots + t^n\phi(n) + \ldots,$$

it hereby appears that we have

$$\theta = 1 + t\theta^4;$$

or writing this for a moment $\theta = u + t\theta^4$, and expanding by Lagrange's theorem, but putting finally $u=1$, we have the value of θ, that is of the generating function,

$$= 1 + [4]^0\,\frac{t}{1} + [8]^1\,\frac{t^2}{1\cdot 2} + [12]^2\,\frac{t^3}{1\cdot 2\cdot 3} + \ldots + [4n]^{n-1}\,\frac{t^n}{1\cdot 2\cdot \ldots \cdot n} + \ldots$$

$$= 1 + t + 4t^2 + 22t^3 + 140t^4 + \ldots,$$

that is,

$$\phi(1) = 1, \quad \phi(2) = 4, \quad \phi(3) = 22, \quad \phi(4) = 140, \ldots$$

and generally

$$\phi(n) = \frac{[4n]^{n-1}}{[n]^n} = \frac{4n\cdot 4n-1\cdot \ldots \cdot 3n+2}{2\cdot 3\cdot \ldots \cdot n}.$$

The results are easily verified for the successive particular cases; thus $n=1$, the points are $1,2,3,4$, and the P- and Q-branches respectively are 13, 24: $\phi(1)=1$. Again $n=2$, the points are $1,2,3,4,5,6,7,8$: we may join $13,24$ or $13,28$ or $17,28$ or $17,68$, leaving in each case four contiguous numbers which may be joined in a single manner: that is, $\phi(2)=4$. Or, what is the same thing, the partitions of 1 are $0001, 0010, 0100, 1000$, whence $\phi(2) = 4\{\phi(0)\}^3\phi(1) = 4$. Again $n=3$, the partitions of 2 are 0002, &c. (4 of this form) and 1100 (6 of this form): that is, $\phi(3) = 4\{\phi(0)\}^3\phi(2) + 6\{\phi(0)\}^2\{\phi(1)\}^2 = 4\cdot 4 + 6\cdot 1 = 22$, and so on.

38. Starting from the $4n$ points P and Q, and joining them in any manner subject to the foregoing conditions, we have a diagram representing two rhizic curves: and colouring the regions we verify that the n roots are all of them right-handed. We have for instance the annexed figure ($n=3$).

[Figure: a circular diagram with $4n=12$ alternating P- and Q-points around the boundary; three P-branches and three Q-branches drawn inside, each P-branch crossing exactly one Q-branch, producing three right-handed intersections.]

Having drawn such a figure we may, by a continuous variation of the several lines, in a variety of ways introduce a double point in the P-curve, or in the Q-curve: and by a continued repetition of the process introduce double points in each or either curve: thus for instance we may from the last figure derive a new figure in which the P-curve has a node at N. It will be observed that here it is no longer the case that each P-branch intersects one and only one Q-branch: the P-branch $1-9$ does not meet any Q-branch, but the P-branch $7-11$ meets two Q-branches. But looking at the figure in a different manner, and considering the P-branches through N as

[p. 38] being either $11-N-1$ and $7-N-9$, or $1-N-7$ and $9-N-11$, then in either case each P-branch intersects one and only one Q-branch: and in this way, in a

[Figure: a second circular diagram in which the P-curve has been deformed so as to acquire a node at the interior point N, with the P-branches passing through N shown in two pairings.]

[Figure: a third related diagram in the same style, illustrating a further continuous variation in which the curves acquire additional double points.]

diagram in which the two curves have each or either of them double points, but neither curve passes through a double point of the other curve, the theorem may be regarded as remaining true—we in fact consider the diagram as the limit of a diagram wherein the curves have no double points. It will be recollected that, the equation $F(z)$ being without equal roots, we cannot have either curve passing through a multiple point of the other curve. And we thus see that the various figures drawn as above without double points are, so to speak, the types of all the different forms of a system of rhizic curves $P = 0$, $Q = 0$.

[p. 39] In connexion with the present paper I give the following list of Memoirs:

Cauchy. Calcul des Indices des fonctions. *Jour. de l'École Polyt. t.* XV. *(1837), pp. 176–229.*—First part seems to have been written in 1833: second part is dated 20th June, 1837. Refers to a memoir presented to the Academy of Turin the 17th Nov. 1831, wherein the principles of the "Calcul des Indices des fonctions" are deduced from the theory of definite integrals: I have not seen this.

Sturm and Liouville. Démonstration d'un théorème de M. Cauchy relatif aux racines imaginaires des équations. *Liouv. t.* I. *(1836), pp. 278–289.*

Sturm. Autres démonstrations du même théorème. *Liouv. t.* I. *(1836), pp. 290–308.*

These two papers contain proofs of the particular theorem relating to the roots of an equation $F(z) = 0$, but do not refer to the general theorem relating to the intersection of the two curves $P = 0$, $Q = 0$: the special theorem of the existence of the n roots of the equation $F(z) = 0$ is considered.

Sylvester. A theory of the syzygetic relations of two rational integral functions, comprising an application to the theory of Sturm's functions and that of the greatest algebraical common measure. *Phil. Trans. t.* CXLIII. *(1853), pp. 407–548.*

De Morgan. A proof of the existence of a root in every algebraic equation, with an examination and extension of Cauchy's theorem on imaginary roots, and remarks on the proofs of the existence of roots given by Argand and Mourey. *Camb. Phil. Trans. t.* X. *(1858), pp. 261–270.*

Contains the important remark that the two curves $P = 0$, $Q = 0$ are such that two branches, one of each curve, cannot inclose a space; also that the two curves always [i.e. at a simple intersection] intersect orthogonally.

Airy, G. B. Suggestion of a proof of the theorem that every algebraic equation has a root. *Camb. Phil. Trans. t.* X. *(1859), pp. 283–289.*

Cayley, A. Sketch of a proof of the theorem that every algebraic equation has a root. *Phil. Mag. t.* XVIII. *(1859), [248], pp. 436–439.*

Walton, W. On a theorem in maxima and minima. *Quart. Math. Jour. t.* X. *(1870), pp. 253–262.* **Cayley, A.** Addition thereto, [562], pp. 262, 263. (Relates to the curves $P = 0$, $Q = 0$.)

Walton, W. Note on rhizic curves. *Quart. Math. Jour. t.* XI. *(1871), pp. 91–98.* First use of the term "rhizic curves": relates chiefly to the configuration of each curve at a multiple point, and of the two at a common multiple point.

Walton, W. On the spoke-asymptotes of rhizic curves. *Quart. Math. Jour. t.* XI. *(1871), pp. 200–202.*

Walton, W. On a property of the curvature of rhizic curves at multiple points. *Quart. Math. Jour. t.* XI. *(1871), pp. 274–281.*

Björling. Sur la séparation des racines d'équations algébriques. *Upsala, Nova Acta Soc. Sci. (1870), pp. 1–35.* (Contains delineations of some rhizic curves.)

562. [Addition to Mr Walton's Paper "On a Theorem in Maxima and Minima."]

[From the Quarterly Journal of Pure and Applied Mathematics, vol. X. (1870), pp. 262, 263.]

[p. 40] In what follows I write x, y, z in place of Mr Walton's u, v, w: (so that if $i = \sqrt{(-1)}$, as usual, we have

$$f(x+iy) = P + iQ);$$

and I attend exclusively to the case where the second differential coefficients of P, Q do not vanish.

There are not on the surface $z = P$ any proper maxima or minima, but only level points, such as at the top of a pass: say there are not any summits or imits, but only cruxes; and moreover at any crux, the two crucial (or level) directions intersect at right angles. Every node of the curve $Q = 0$ is subjacent to a crux of the surface $z = P$: and moreover the two directions of the curve $Q = 0$ at the node are at right angles to each other; hence, considering the intersection of the surface $z = P$ by the cylinder $Q = 0$, the path $Q = 0$ on the surface has a node at the crux; or say there are at the crux two directions of the path; these cross at right angles, and are consequently separated the one from the other by the crucial directions; that is to say, there is one path ascending, and another path descending, each way from the crux. And the complete statement is: that the elevation of the path is then only a maximum or minimum when the path passes through a crux; and that at any crux there are two paths, one ascending, the other descending, each way from the crux.

The analytical demonstration is exceeding simple; we have

$$\left(\frac{dP}{dy} + i\frac{dQ}{dy}\right) = i\left(\frac{dP}{dx} + i\frac{dQ}{dx}\right);$$

[p. 41] that is,

$$\frac{dP}{dy} = -\frac{dQ}{dx}, \qquad \frac{dQ}{dy} = \frac{dP}{dx},$$

and passing thence to the second differential coefficients, we may write

$$\frac{dP}{dx} = \frac{dQ}{dy} = L, \qquad \frac{dP}{dy} = -\frac{dQ}{dx} = M,$$

$$\frac{d^2P}{dxdy} = -\frac{d^2Q}{dx^2} = \frac{d^2Q}{dy^2} = a,$$

$$\frac{d^2Q}{dxdy} = \frac{d^2P}{dx^2} = -\frac{d^2P}{dy^2} = b,$$

so that we have

$$\delta P = L\,\delta x + M\,\delta y, \qquad \delta Q = -M\,\delta x + L\,\delta y,$$

$$\delta^2 P = (b,\, a,\, -b \mid \delta x,\, \delta y)^2, \qquad \delta^2 Q = (-a,\, b,\, a \mid \delta x,\, \delta y)^2.$$

Hence, for the maximum or minimum elevation of the path, we have $0 = \delta P$, where $\delta Q = 0$; that is, $0 = -\frac{L^2+M^2}{L}\delta x$, and therefore $L^2 + M^2 = 0$; that is, $L = 0$, $M = 0$; and at any such point $\delta z = 0$, that is, there is a crux of the surface $z = P$; and $\delta Q = 0$, that is, there is a node of the curve $Q = 0$. Moreover the crucial directions for the surface $z = P$ are given by the equation $(b,\, a,\, -b \mid \delta x,\, \delta y)^2 = 0$, or these are at right angles to each other; and the nodal directions for the curve $Q = 0$ are given by $(-a,\, b,\, a \mid \delta x,\, \delta y)^2 = 0$, or these are likewise at right angles to each other.

C. IX.

563. Note on the Transformation of Two Simultaneous Equations

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XI. *(1871), pp. 266, 267.]*

[p. 42] Writing in Mr Walton's equations (1) and (2)

$$\frac{a}{\delta},\ \frac{b}{\delta},\ \frac{c}{\delta},\ \frac{\alpha}{\delta},\ \frac{\beta}{\delta},\ \frac{\gamma}{\delta}$$

instead of a, b, c, α, β, γ respectively; and putting for shortness

$$\begin{aligned} A &= b\gamma - c\beta, & F &= a\delta - d\alpha, \\ B &= c\alpha - a\gamma, & G &= b\delta - d\beta, \\ C &= a\beta - b\alpha, & H &= c\delta - d\gamma, \end{aligned}$$

the equations become

$$\frac{a(b-c)}{F} + \frac{b(c-a)}{G} + \frac{c(a-b)}{H} = 0,$$

$$\frac{\alpha(\beta-\gamma)}{F} + \frac{\beta(\gamma-\alpha)}{G} + \frac{\gamma(\alpha-\beta)}{H} = 0.$$

Multiplying by FGH and effecting some obvious transformations, the equations become

$$aAF + bBG + cCH = 0, \qquad \alpha AF + \beta BG + \gamma CH = 0 \tag{18}$$

whence also

$$AF^2 + BG^2 + CH^2 = 0. \tag{19}$$

Now regarding $(\alpha, \beta, \gamma, \delta)$ as the coordinates of a point in space, the equations (18) and (19) represent each of them a cone having for vertex the point $\alpha : \beta : \gamma : \delta = a : b : c : d$, viz. (18) is a quadric cone, (19) a cubic cone; they intersect therefore in six lines; and it may be shown that these are

the line	$\alpha : \beta : \gamma = a : b : c$	(twice) 2
"	$\beta : \gamma : \delta = b : c : d$	1
"	$\gamma : \alpha : \delta = c : a : d$	1
"	$\alpha : \beta : \delta = a : b : d$	1
"	$\beta - \gamma : \gamma - \alpha : \alpha - \beta : \delta = b - c : c - a : a - b : d$	1

agreeing with Mr Walton's result.

564. On a Theorem in Elimination

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XII. *(1873), pp. 5, 6.]*

[p. 43] I find among my papers the following example of a theorem in elimination communicated to me by Prof. Sylvester. Writing

$$\begin{aligned} \phi &= ax^3 + 3bx^2y + 3cxy^2 + dy^3, \\ \phi_1 &= bx^2 + 2cxy + dy^2, \\ \phi_2 &= cx + dy, \\ \phi_3 &= d\,; \end{aligned}$$

$$\begin{aligned} f &= bx^3 + 3cx^2y + 3dxy^2 + ey^3, \\ f_1 &= cx^2 + 2dxy + ey^2, \\ f_2 &= dx + ey, \\ f_3 &= e\,; \end{aligned}$$

then we have

$$\Delta_a \cdot R(f,\phi) = \Delta f \cdot R(\phi_1, f_1)^2 \cdot R(\phi_2, f_2)^2,$$

viz. $R(f,\phi)$ is the resultant of the functions (f,ϕ), and similarly $R(\phi_1, f_1)$, $R(\phi_2, f_2)$. Moreover, Δf is the discriminant of f; and $\Delta_a R(f,\phi)$ is the discriminant of $R(f,\phi)$ in regard to a. The equation thus is

$$\Delta_a\big[(ae-4bd+3c^2)^3 - 27(ace-ad^2-b^2e-c^3+2bcd)^2\big]$$
$$= (b^2e^2+4bd^3+4c^3e-3c^2d^2-6bcde)^2\,(d^2-2cde+be^2)^2$$

or, what is the same thing, reversing the order of the letters (a,b,c,d,e), it is

$$\Delta_e\big[(ae-4bd+3c^2)^3 - 27(ace-ad^2-b^2e-c^3+2bcd)^2\big]$$
$$= (a^2d^2+4ac^3+4b^3d-3b^2c^2-6abcd)^2\,(b^2-2abc+a^2d)^2,$$

[p. 44] viz. arranging in powers of e, the function is

$$\begin{aligned}
&e^3\cdot a^3\\
&+3e^2\cdot -a^2(4bd-3c^2)-9(ac-b^2)^2\\
&+3e\cdot a(4bd-3c^2)^2+18(ac-b^2)(ad^2-2bcd+c^3)\\
&+1\cdot -(4bd-3c^2)^3-27(ad^2-2bcd+c^3)^2,
\end{aligned}$$

which last coefficient is

$$= -d^2(27a^2d^2+54ac^3+64b^3d-36b^2c^2-108abcd),$$

and the discriminant of this cubic function of e is

$$= (a^2d^2+4ac^3+4b^3d-3b^2c^2-6abcd)^2\,(b^2-2abc+a^2d)^2.$$

The occurrence of the factor

$$a^2d^2+4ac^3+4b^3d-3b^2c^2-6abcd$$

is accounted for as the resultant in regard to e of the invariants I, J; we, in fact, have

$$(ac-b^2)I-aJ = -(ac-b^2)(-4bd+3c^2)-a(-ad^2-c^3+2bcd)$$
$$= a^2d^2+4ac^3+4b^3d-3b^2c^2-6abcd,$$

and the identity itself may be proved without any particular difficulty.

565. Note on the Cartesian

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XII. (1873), pp. 16–19.]

[p. 45] The following are doubtless known theorems, but the form of statement, and the demonstration of one of them, may be interesting.

A point P on a Cartesian has three "opposite" points on the curve, viz. if the axial foci are A, B, C, then the opposite points are P_a, P_b, P_c where

P_a is intersection of line	PA	with circle	PBC,
P_b "	PB	"	PCA,
P_c "	PC	"	PAB.

And, moreover, supposing in the three circles respectively, the diameters at right angles to PA, PB, PC are $\alpha\alpha'$, $\beta\beta'$, $\gamma\gamma'$ respectively, then the points α, α', β, β', γ, γ' lie by threes in two lines passing through P, viz. one of these, say $P\alpha\beta\gamma$, is the tangent, and the other $P\alpha'\beta'\gamma'$ the normal, at P; and then the tangents and normals at the opposite points are $P_a\alpha$ and $P_a\alpha'$, $P_b\beta$ and $P_b\beta'$, $P_c\gamma$, and $P_c\gamma'$ respectively.

There exists a second Cartesian with the same axial foci A, B, C, and passing through the points P, P_a, P_b, P_c (which are obviously opposite points in regard thereto); the tangent at P is $P\alpha'\beta'\gamma'$ and the normal is $P\alpha\beta\gamma$; and the tangent and the normal at the other points are $P_a\alpha'$ and $P_a\alpha$, $P_b\beta'$ and $P_b\beta$, $P_c\gamma'$ and $P_c\gamma$ respectively: viz. the two curves cut at right angles at each of the four points.

Starting with the foci A, B, C and the point P, the points P_a, P_b, P_c are constructed as above, without the employment of the Cartesian; there are through P with the foci A, B, C two and only two Cartesians; and if it is shown that these pass through one of the opposite points, say P_b, they must, it is clear, pass through

[p. 46] the other two points P_a, P_c. I propose to find the two Cartesians in question. To fix the ideas, let the points C, B, A be situate in order as shown in the figure, their distances from a fixed point O being a, b, c, so that writing α, β, $\gamma = b-c$, $c-a$, $a-b$ respectively, we have $\alpha+\beta+\gamma=0$, and α, γ will represent the positive distances CB and BA respectively, and $-\beta$ the positive distance AC. Suppose, moreover, that the distances PA, PB, PC regarded as positive are R, S, T respectively; and that the distances P_bA, P_bB, P_bC regarded as positive are R', S', T' respectively.

[Figure: a Cartesian curve with axial foci A, B, C marked along a horizontal line; points P and P_b shown on the curve with the various distances and the diameters $\alpha\alpha'$, $\beta\beta'$, $\gamma\gamma'$ of the three opposite-point circles indicated.]

Suppose that for a current point Q the distances QA, QB, QC regarded as indifferently positive, or negative, are r, s, t respectively; then the equation of a bicircular quartic having the points A, B, C for axial foci is

$$lr+ms+nt=0,$$

where l, m, n are constants; and this will be a Cartesian if only

$$\frac{l^2}{\alpha}+\frac{m^2}{\beta}+\frac{n^2}{\gamma}=0.$$

We have the same curve whatever be the signs of l, m, n, and hence making the curve pass through P, we may, without loss of generality, write

$$lR+mS+nT=0,$$

R, S, T denoting the positive distances PA, PB, PC as above. We have thus for the ratios $l:m:n$, two equations, one simple, the other quadric; and there are thus two systems of values, that is, two Cartesians with the foci A, B, C, and passing through P.

I proceed to show that for one of these we have $-lR'+mS'+nT'=0$, and for the other $lR'+mS'-nT'=0$, or, what is the same thing, that the values of $l:m:n$ are

$$l:m:n=-(ST'+S'T):TR'+T'R:RS'-R'S,$$

and

$$l:m:n=(ST'-S'T):-(TR'+T'R):RS'+R'S\,;$$

viz. that the equations of the two Cartesians are

$$\begin{vmatrix} r & s & t \\ R & S & T \\ -R' & S' & T' \end{vmatrix}=0, \qquad \text{and} \qquad \begin{vmatrix} r & s & t \\ R & S & T \\ R' & S' & -T' \end{vmatrix}=0,$$

[p. 47] respectively; this being so each of the Cartesians will, it is clear, pass through the point P_b, and therefore also through P_a and P_c.

The geometrical relations of the figure give

$$\begin{aligned}
\alpha R^2 + \beta S^2 + \gamma T^2 &= -\alpha\beta\gamma,\\
\alpha R'^2 + \beta S'^2 + \gamma T'^2 &= -\alpha\beta\gamma,\\
RT' + R'T &= -\beta(S+S'),\\
\gamma\alpha &= SS',\\
\gamma\, TT' &= \alpha\, RR',
\end{aligned}$$

to which might be joined

$$\begin{aligned}
R'^2 S + \alpha\beta(S+S') + RS'^2 &= SS'(S+S'),\\
T'^2 S + \alpha\beta(S+S') + T^2 S' &= SS'(S+S'),\\
SRT' &= S'RT,\\
ST'R &= S'TR,
\end{aligned}$$

but these are not required for the present purpose.

As regards the first Cartesian, we have to verify that

$$\frac{(ST'+S'T)^2}{\alpha} + \frac{(TR'+T'R)^2}{\beta} + \frac{(RS'-R'S)^2}{\gamma} = 0.$$

The left-hand side is

$$\frac{S^2T'^2 + S'^2T^2 + 2\gamma\alpha\, TT'}{\alpha} + \frac{\beta^2(S+S')^2 + 2\gamma\alpha\,[\mathit{unreadable}]}{\beta} + \frac{S^2R'^2 + S'^2R^2 - 2\gamma\alpha\, RR'}{\gamma},$$

viz. this is

$$= S^2\left(\frac{T'^2}{\alpha} + \frac{[\mathit{unreadable}]}{\beta} + \frac{R'^2}{\gamma}\right) + S'^2\left(\frac{T^2}{\alpha} + \frac{[\mathit{unreadable}]}{\beta} + \frac{R^2}{\gamma}\right) + 2\alpha\beta\gamma + 2(\gamma TT' - \alpha RR'),$$

which is

$$= S^2\left(\frac{-\alpha\beta\gamma}{\alpha\gamma}\right) + S'^2\left(\frac{-\alpha\beta\gamma}{\alpha\gamma}\right) + 2\alpha\beta\gamma + 2(\gamma TT' - \alpha RR'),$$

and since the first and second terms are together $= -\dfrac{2\beta}{\gamma\alpha}S^2S'^2$, that is, $= -2\alpha\beta\gamma$, the whole is as it should be $= 0$.

In precisely the same manner we have

$$\frac{(ST'-S'T)^2}{\alpha} + \frac{(TR'+T'R)^2}{\beta} + \frac{(RS'+R'S)^2}{\gamma} = 0,$$

which is the condition for the second Cartesian: and the theorem in question is thus proved.

566. On the Transformation of the Equation of a Surface to a Set of Chief Axes

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XII. (1873), pp. 34–38.]

[p. 48] We have at any point P of a surface a set of chief axes (PX, PY, PZ), viz. these are, say the axis of Z in the direction of the normal, and those of X, Y in the directions of the tangents to the two curves of curvature respectively. It may be required to transform the equation of the surface to the axes in question; to show how to effect this, take the original (rectangular) coordinates of the point P, (x, y, z) for the point P, $x+\delta x$, $y+\delta y$, $z+\delta z$ for the like coordinates of any other point on the surface, so that $(\delta x, \delta y, \delta z)$ are the coordinates of the point referred to the origin P; the equation of the surface, writing down only the terms of the first and second orders in the coordinates δx, δy, δz, is

$$A\,\delta x + B\,\delta y + C\,\delta z + \tfrac{1}{2}(a, b, c, f, g, h \mid \delta x, \delta y, \delta z)^2 + \&\text{c.} = 0,$$

where (A, B, C) are the first derived functions and (a, b, c, f, g, h) the second derived functions of U for the values (x, y, z) which belong to the given point P, if $U = 0$ is the equation of the surface in terms of the original coordinates (x, y, z); we have X, Y, Z linear functions of $(\delta x, \delta y, \delta z)$; say

$$\begin{array}{c|ccc} & \delta x & \delta y & \delta z \\ \hline X & \alpha_1 & \beta_1 & \gamma_1 \\ Y & \alpha_2 & \beta_2 & \gamma_2 \\ Z & \alpha & \beta & \gamma \end{array}$$

that is, $X = \alpha_1\delta x + \beta_1\delta y + \gamma_1\delta z$, &c. and $\delta x = \alpha_1 X + \alpha_2 Y + \alpha Z$, &c. where the coefficients satisfy the ordinary relations in the case of transformation between two sets of rectangular axes; and the transformed equation is therefore

$$A(\alpha_1 X + \alpha_2 Y + \alpha Z) + B(\beta_1 X + \beta_2 Y + \beta Z) + C(\gamma_1 X + \gamma_2 Y + \gamma Z)$$
$$+(a, b, c, f, g, h \mid \alpha_1 X + \alpha_2 Y + \alpha Z,\ \beta_1 X + \beta_2 Y + \beta Z,\ \gamma_1 X + \gamma_2 Y + \gamma Z)^2 = 0,$$

[p. 49] or, as this may be written,

$$\begin{aligned} &X(A\alpha_1 + B\beta_1 + C\gamma_1) + Y(A\alpha_2 + B\beta_2 + C\gamma_2) + Z(A\alpha + B\beta + C\gamma) \\ &\quad + \tfrac{1}{2}X^2\,(a, \ldots)(\alpha_1, \beta_1, \gamma_1)^2 \\ &\quad + \tfrac{1}{2}Y^2\,(a, \ldots)(\alpha_2, \beta_2, \gamma_2)^2 \\ &\quad + XY\,(a, \ldots)(\alpha_1, \beta_1, \gamma_1)(\alpha_2, \beta_2, \gamma_2) \\ &\quad + XZ\,(a, \ldots)(\alpha_1, \beta_1, \gamma_1)(\alpha, \beta, \gamma) \\ &\quad + YZ\,(a, \ldots)(\alpha_2, \beta_2, \gamma_2)(\alpha, \beta, \gamma) \\ &\quad + \tfrac{1}{2}Z^2\,(a, \ldots)(\alpha, \beta, \gamma)^2 \quad + \&c. = 0, \end{aligned}$$

where the &c. refers to terms of the form $(X, Y, Z)^3$ and higher powers.

But in order that the new axes may be chief axes, we must have

$$A\alpha_1 + B\beta_1 + C\gamma_1 = 0, \qquad A\alpha_2 + B\beta_2 + C\gamma_2 = 0,$$
$$(a, \ldots)(\alpha_1, \beta_1, \gamma_1)(\alpha_2, \beta_2, \gamma_2) = 0,$$

so that putting for shortness

$$A\alpha + B\beta + C\gamma = V,$$

the equation becomes

$$\begin{aligned} &VZ + \tfrac{1}{2}X^2\,(a, \ldots)(\alpha_1, \beta_1, \gamma_1)^2 + \tfrac{1}{2}Y^2\,(a, \ldots)(\alpha_2, \beta_2, \gamma_2)^2 \\ &\quad + XZ\,(a, \ldots)(\alpha_1, \beta_1, \gamma_1)(\alpha, \beta, \gamma) \\ &\quad + YZ\,(a, \ldots)(\alpha_2, \beta_2, \gamma_2)(\alpha, \beta, \gamma) \\ &\quad + \tfrac{1}{2}Z^2\,(a, \ldots)(\alpha, \beta, \gamma)^2 \quad + \&c. = 0. \end{aligned}$$

We have

$$A : B : C = \beta_1\gamma_2 - \beta_2\gamma_1 : \gamma_1\alpha_2 - \gamma_2\alpha_1 : \alpha_1\beta_2 - \alpha_2\beta_1,$$

that is,

$$A : B : C = \alpha : \beta : \gamma,$$

and thence

$$\alpha, \beta, \gamma = \frac{A}{V}, \frac{B}{V}, \frac{C}{V}, \qquad V = \sqrt{A^2 + B^2 + C^2}.$$

I write

$$\frac{1}{p_1} = (a, \ldots)(\alpha_1, \beta_1, \gamma_1)^2,$$

and also for a moment

$$\frac{V}{p_1} = \mathcal{P}(\alpha_1, \beta_1, \gamma_1),$$
$$Q = (\,h,\ b,\ f\,)(\alpha_1, \beta_1, \gamma_1),$$
$$R = (\,g,\ f,\ c\,)(\alpha_1, \beta_1, \gamma_1).$$

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[p. 50] We find

$$P\alpha_1 + Q\beta_1 + R\gamma_1 = (a, \ldots)(\alpha_1, \beta_1, \gamma_1)^2 - \frac{V}{p_1} = 0,$$

$$P\alpha_2 + Q\beta_2 + R\gamma_2 = (a, \ldots)(\alpha_1, \beta_1, \gamma_1)(\alpha_2, \beta_2, \gamma_2) - \frac{V}{p_1}(\alpha_1\alpha_2 + \beta_1\beta_2 + \gamma_1\gamma_2) = 0,$$

and thence

$$P : Q : R = \beta_1\gamma_2 - \beta_2\gamma_1 : \gamma_1\alpha_2 - \gamma_2\alpha_1 : \alpha_1\beta_2 - \alpha_2\beta_1,$$
$$= \alpha : \beta : \gamma,$$

or say

$$P, Q, R = \theta_1 A,\ \theta_1 B,\ \theta_1 C\,;$$

we have thus the equations

$$\left(a - \frac{1}{p_1}\right)\alpha_1 + h\beta_1 + g\gamma_1 = \theta_1 A,$$

$$h\alpha_1 + \left(b - \frac{1}{p_1}\right)\beta_1 + f\gamma_1 = \theta_1 B,$$

$$g\alpha_1 + f\beta_1 + \left(c - \frac{1}{p_1}\right)\gamma_1 = \theta_1 C,$$

and joining hereto

$$(A, B, C)(\alpha_1, \beta_1, \gamma_1) = 0,$$

we eliminate α_1, β_1, γ_1 and obtain the equation

$$\begin{vmatrix} a - \dfrac{1}{p_1} & h & g & A \\ h & b - \dfrac{1}{p_1} & f & B \\ g & f & c - \dfrac{1}{p_1} & C \\ A & B & C & 0 \end{vmatrix} = 0.$$

and in like manner writing

$$\frac{1}{p_2} = (a, \ldots)(\alpha_2, \beta_2, \gamma_2)^2,$$

we have the same equation for p_2; wherefore p_1, p_2 are the roots of the quadric equation

$$\begin{vmatrix} a - \dfrac{1}{p} & h & g & A \\ h & b - \dfrac{1}{p} & f & B \\ g & f & c - \dfrac{1}{p} & C \\ A & B & C & 0 \end{vmatrix} = 0.$$

[p. 51] Moreover, p_1, p_2 being thus determined, we have α_1, β_1, γ_1, θ_1 proportional to the determinants formed with the matrix

$$\begin{vmatrix} a-\dfrac{1}{p_1} & h & g & A \\ h & b-\dfrac{1}{p_1} & f & B \\ g & f & c-\dfrac{1}{p_1} & C \end{vmatrix},$$

say, $\alpha_1, \beta_1, \gamma_1, \theta_1 = k\mathfrak{A}_1, k\mathfrak{B}_1, k\mathfrak{C}_1, k\Omega_1$, where $\mathfrak{A}_1$, $\mathfrak{B}_1$, $\mathfrak{C}_1$, Ω_1 are the determinants in question; and then $1 = k^2(\mathfrak{A}_1^2 + \mathfrak{B}_1^2 + \mathfrak{C}_1^2)$, or we have

$$\theta_1 = \frac{\Omega_1}{\sqrt{\mathfrak{A}_1^2 + \mathfrak{B}_1^2 + \mathfrak{C}_1^2}}.$$

But we find at once

$$(a, \ldots)(\alpha_1, \beta_1, \gamma_1)(\alpha, \beta, \gamma) = \theta_1(A\alpha + B\beta + C\gamma) = \theta_1 V,$$

that is,

$$(a, \ldots)(\alpha_1, \beta_1, \gamma_1)(\alpha, \beta, \gamma) = \frac{V\Omega_1}{\sqrt{\mathfrak{A}_1^2 + \mathfrak{B}_1^2 + \mathfrak{C}_1^2}},$$

and in the same manner

$$(a, \ldots)(\alpha_2, \beta_2, \gamma_2)(\alpha, \beta, \gamma) = \frac{V\Omega_2}{\sqrt{\mathfrak{A}_2^2 + \mathfrak{B}_2^2 + \mathfrak{C}_2^2}}.$$

Hence the transformed equation is

$$VZ + \tfrac{1}{2}\frac{X^2}{p_1} + \tfrac{1}{2}\frac{Y^2}{p_2} + XZ\,\frac{V\Omega_1}{\sqrt{\mathfrak{A}_1^2 + \mathfrak{B}_1^2 + \mathfrak{C}_1^2}} + YZ\,\frac{V\Omega_2}{\sqrt{\mathfrak{A}_2^2 + \mathfrak{B}_2^2 + \mathfrak{C}_2^2}} + \tfrac{1}{2}Z^2\,(a, \ldots)(\alpha, \beta, \gamma)^2 + \text{\&c.} = 0,$$

where it will be recollected that $V = \sqrt{A^2 + B^2 + C^2}$. The &c. refers as before to the terms $(X, Y, Z)^3$ and higher powers, which are obtained from the corresponding terms in δx, δy, δz by substituting for these their values $\delta x = \alpha_1 X + \alpha_2 Y + \alpha Z$, &c., where the coefficients have the values above obtained for them. It will be observed, that the radii of curvature are Vp_1, Vp_2, and that the process includes an investigation of the values of these radii of curvature similar to the ordinary one; the novelty is in the terms in XZ, YZ, and Z^2. But regarding X, Y as small quantities of the first order, Z is of the second order, and the terms in XZ, YZ are of the third order, and that in Z^2 of the fourth order.

567. On an Identical Equation Connected with the Theory of Invariants

[From the Quarterly Journal of Pure and Applied Mathematics, vol. XII. (1873), pp. 115–118.]

[p. 52] Write

$$\begin{aligned} a &= g - h, \\ b &= h - f, \\ c &= f - g, \end{aligned}$$

equations implying a fourth equation forming with them the system

$$\begin{aligned} \cdot\ -h + g - a &= 0, \\ h\ \cdot\ -f - b &= 0, \\ -\,g + f\ \cdot\ -c &= 0, \\ a + b + c\,\cdot\ &= 0, \end{aligned}$$

and also

$$af + bg + ch = 0.$$

Then, putting for shortness

$$\begin{aligned}
P &= (bg - ch)(ch - af)(af - bg), \\
Q &= a^2g^2h^2 + b^2h^2f^2 + c^2f^2g^2 + a^2b^2c^2, \\
R &= a^2f^2(a^2 + f^2) + b^2g^2(b^2 + g^2) + c^2h^2(c^2 + h^2),
\end{aligned}$$

we have

$$2P + Q - R = 0,$$

viz. substituting for a, b, c their values $g - h$, $h - f$, $f - g$, this is an identical equation.

[p. 53] The direct verification is however somewhat tedious, and the equation may be proved more easily as follows:

In the terms $a^2 + f^2$, $b^2 + g^2$, $c^2 + h^2$ of R, substituting for a, b, c their values, we find

$$R = (f^2 + g^2 + h^2)(a^2f^2 + b^2g^2 + c^2h^2) - 2fgh(a^2f + b^2g + c^2h),$$

which may be written

$$R = -2(f^2 + g^2 + h^2)(bcgh + cahf + abfg) - 2fgh(a^2f + b^2g + c^2h).$$

We have then

$$2P = -2bcgh(bg - ch) - 2cahf(ch - af) - 2abfg(af - bg),$$

and thence

$$\begin{aligned}
2P - R = 2bcgh&(f^2 + g^2 + h^2 - bg + ch) \\
&+ 2cahf(f^2 + g^2 + h^2 - ch + af) \\
&+ 2abfg(f^2 + g^2 + h^2 - af + bg) \\
&+ 2fgh(a^2f + b^2g + c^2h),
\end{aligned}$$

which is at once converted into

$$\begin{aligned}
2P - R = 2bcgh&\{a^2 + f(f + g + h)\} \\
&+ 2cahf\{b^2 + g(f + g + h)\} \\
&+ 2abfg\{c^2 + h(f + g + h)\} \\
&+ 2fgh(a^2f + b^2g + c^2h)\,;
\end{aligned}$$

or, what is the same thing,

$$2P - R = 2fgh\big\{(bc + ca + ab)(f + g + h) + a^2f + b^2g + c^2h\big\} + 2abc(agh + bhf + cfg),$$

where, since

$$agh + bhf + cfg = -abc,$$

the last term is

$$= -2a^2b^2c^2.$$

But from the equation last written down we deduce at once

$$Q = 2a^2b^2c^2 - 2fgh(bcf + cag + abh),$$

and we thence have

$$2P + Q - R = 2fgh\big\{(bc + ca + ab)(f + g + h) + (a^2f + b^2g + c^2h) - bcf - cag - abh\big\},$$

which is

$$= 2fgh(a + b + c)(af + bg + ch),$$

and consequently $= 0$, the theorem in question.

[p. 54] Instead of a, b, c, f, g, h, I write $aW \div YZ$, $bW \div ZX$, $cW \div XY$, $f \div X$, $g \div Y$, $h \div Z$: we have therefore

$$\begin{aligned}&\cdot\ -hY + gZ - aW = 0,\\ &hX\ \cdot\ -fZ - bW = 0,\\ &-gX + fY\ \cdot\ -cW = 0,\\ &aX + bY + cZ\ \cdot\ = 0,\end{aligned}$$

and as before

$$af + bg + ch = 0.$$

Moreover, omitting a common factor, the new values of P, Q, R are

$$\begin{aligned}&P = XYZW(bg - ch)(ch - af)(af - bg),\\ &Q = a^2g^2h^2X^4 + b^2h^2f^2Y^4 + c^2f^2g^2Z^4 + a^2b^2c^2W^4,\\ &R = a^2f^2(a^2X^2W^2 + f^2Y^2Z^2) + b^2g^2(b^2Y^2W^2 + g^2Z^2X^2) + c^2h^2(c^2Z^2W^2 + h^2X^2Y^2),\end{aligned}$$

and the identical equation is, as before,

$$2P + Q - R = 0.$$

Consider the operative symbols

$$d_{x_1},\ d_{y_1},\ d_{x_2},\ d_{y_2},$$

$$d_{x_3},\ d_{y_3},\ d_{x_4},\ d_{y_4},$$

and write $a = d_{x_1}d_{y_2} - d_{y_1}d_{x_2} = 12$, &c., that is

$$\begin{aligned}a &= 23, & f &= 14,\\ b &= 31, & g &= 24,\\ c &= 12, & h &= 34,\end{aligned}$$

and also $X = x\,d_{x_1} + y\,d_{y_1}$, &c. say

$$X = \nabla_1,\ Y = \nabla_2,\ Z = \nabla_3,\ W = \nabla_4.$$

These values of a, b, c, f, g, h, X, Y, Z, W satisfy the above written equations of connexion, and therefore the identical equation $2P + Q - R = 0$. Hence taking U to denote the quartic function $U = (a, b, c, d, e)(x, y)^4$, and therefore $U_1 = (a, \ldots)(x_1, y_1)^4$, &c., we have

$$(2P + Q - R)\,U_1U_2U_3U_4 = 0,$$

where, after the differentiations, $(x_1, y_1) \ldots (x_4, y_4)$ are to be each of them replaced by (x, y).

Observe that P is the sum of three positive and three negative terms, but that after the omission of the suffixes each term taken with its proper sign becomes equal to the same quantity, and the value of P is $= 6$ times any one term thereof. Thus omitting for the moment the factor $\nabla_1\nabla_2\nabla_3\nabla_4$, two of the terms are $-(af)^2bg + af(bg)^2$, that is,

$$-(14 \cdot 23)^2(24 \cdot 31) + (14 \cdot 23)(24 \cdot 31)^2,$$

[p. 55] and, if in the first term we interchange 3 and 4, it becomes $-(13 \cdot 24)^2(23 \cdot 41)$, that is, $+(14 \cdot 23)(24 \cdot 31)^2$, viz. it becomes equal to the second term. As regards Q the terms are all positive and become equal to each other; and the like as regards R: hence we have

$$\left\{12\,\nabla_1\nabla_2\nabla_3\nabla_4(14 \cdot 23)(24 \cdot 31)^2 + 4\nabla_1^2(23)^2(34)^2(42)^2 - 6\nabla_1^2\nabla_2^2(43)^2(14)^2\right\} U_1U_2U_3U_4 = 0,$$

which, omitting a numerical factor $6 \cdot 2 \cdot 12^2 \cdot 2 \cdot 24^2 \cdot 4 = 3^2 \cdot 2^{15}$, is in fact the well-known equation

$$\Omega + JU - IH = 0,$$

where

$$
\begin{aligned}
U &= (a, b, c, d, e)(x, y)^4, \\
\Omega &= \text{Disc.}(ax + by, bx + cy, cx + dy, dx + ey)(\xi, \eta)^3, \\
&= (ax + by)^3(dx + ey)^2 + \&\text{c.}, \\
I &= ae - 4bd + 3c^2, \\
J &= ace - ad^2 - b^2e - c^3 + 2bcd,
\end{aligned}
$$

viz. attending only to the coefficient of x^6, this equation is

$$a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd + a(ace - ad^2 - b^2e - c^3 + 2bcd) + (ac - b^2)(ae - 4bd + 3c^2) = 0.$$

[Continued on following pages.]